

Regime 45: Hidden Markov / Latent State Q4

EURUSD · 60-minute bars (H1) · n=25,000 bars

Symbol: EURUSD

Bar timeframe: 60 minutes (H1)

Total bars in backtest sample: 25,000

Bars with this regime (primary label): 163 (0.65% of sample)

Quadrant: Q4

Win rates (wr_1r-wr_4r): fraction of trades where price reached $\geq 1R \dots \geq 4R$ favorable excursion before ATR-based stop (see scorecard script).

rank_score: blends wr_1r-wr_4r and lightly penalizes strategies with trades < 8 .

Next pages: Regime52Params (classifier + indicator windows), this regime's taxonomy line, then each strategy's name, signal_kind, side rule, and numeric thresholds.

Regime detection — parameter terms and values (Regime52Params)

EURUSD · 60-minute bars (H1) · n=25,000 bars

Regime52Params — shared by the 52-regime classifier and signal indicators:

```
adx_period: 14
rsi_period: 14
ema_slow: 50
ema_ob: 20
ema_fast: 12
structure_lb: 10
mom_lookback: 60
bb_period: 20
z_period: 20
squeeze_win: 250
vol_cluster_win: 48
hmm_vol_quantile: 0.75
```

Classifier rules (thresholds per regime id, claim order, optional ctx_* columns) live in forex_regime/regimes52/classify.py — add context columns to activate macro/event regimes when those rows have data.

This regime (id 45) — taxonomy terms
EURUSD · 60-minute bars (H1) · n=25,000 bars

Section: SECTION 8 — Microstructure

Narrative / rule summary:
Hidden Markov Regime — latent discrete states drive parameters (Hamilton-style switching)

Strategies for regime 45 — names, keys, sides, signal values

EURUSD · 60-minute bars (H1) · n=25,000 bars

Sample: EURUSD · 60-minute bars (H1) · n=25,000 bars

Strategy rows for this regime (terms). Entries are counted only when regime52_id matches this regime; signal math is shared across all regimes in the same quadrant— only labels differ.

- R45-S01: Liquidity sweep fade — Hidden Markov / Latent State
signal_kind='wick_reversal'; side=+1 → only bullish rule arm fires long
- R45-S02: Vol-cluster straddle / flat gamma proxy fade — Hidden Markov / Latent State
signal_kind='vol_spike_fade'; side=-1 → only bearish rule arm fires short
- R45-S03: Event reversal template — Hidden Markov / Latent State
signal_kind='narrow_range_break_fake'; side=+1 → only bullish rule arm fires long
- R45-S04: Regime-switch cautious drift — Hidden Markov / Latent State
signal_kind='tiny_drift'; side=+1 → only bullish rule arm fires long

Signal kinds — numeric values and windows (same for every regime using that kind):

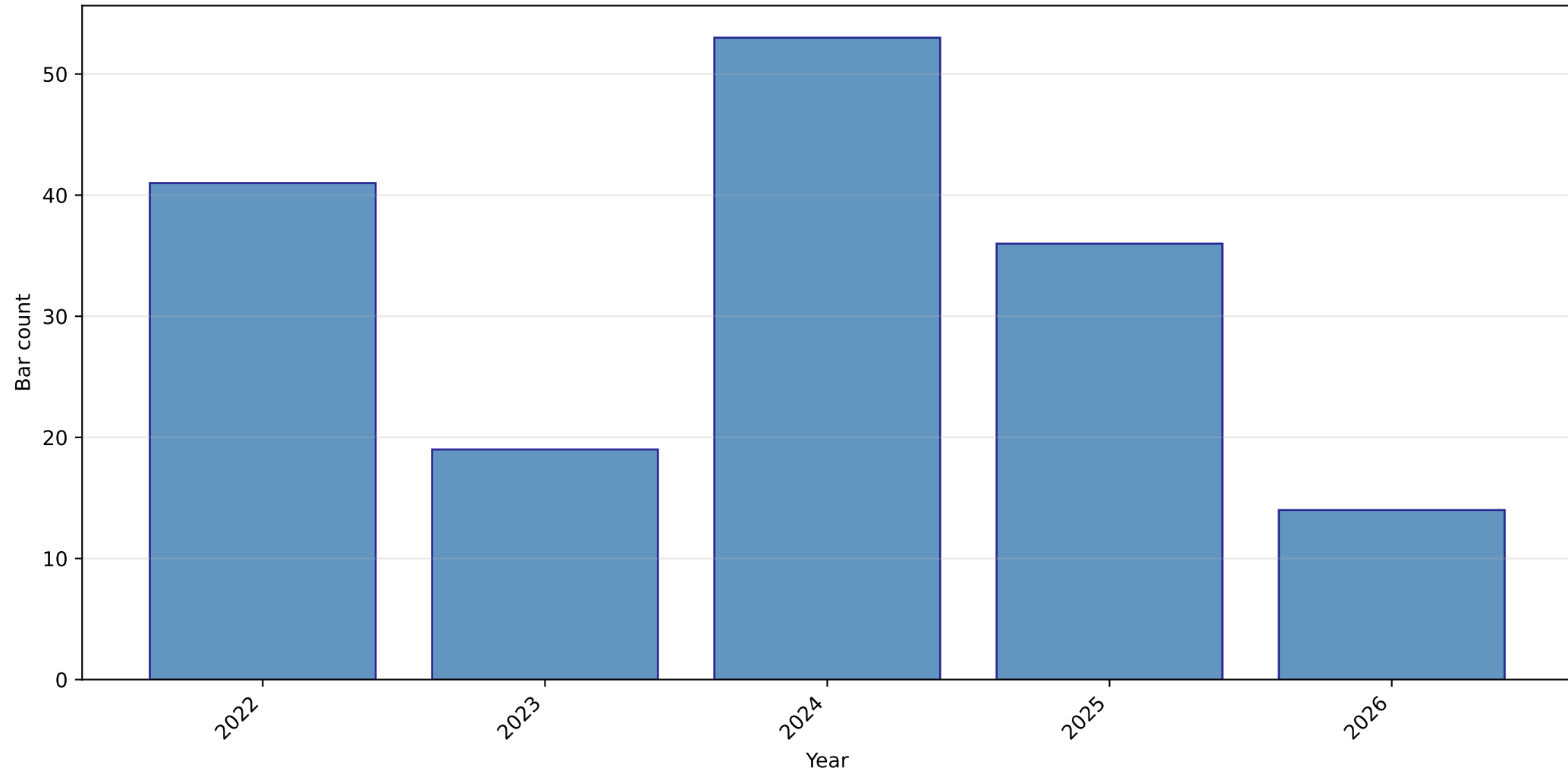
[wick_reversal]
Lower wick > 1.2×|body| & bullish bar → long; upper wick > 1.2×|body| & bearish bar → short

[vol_spike_fade]
ATR(14) > 1.55× 50-bar ATR mean → fade: long on bearish close, short on bullish close

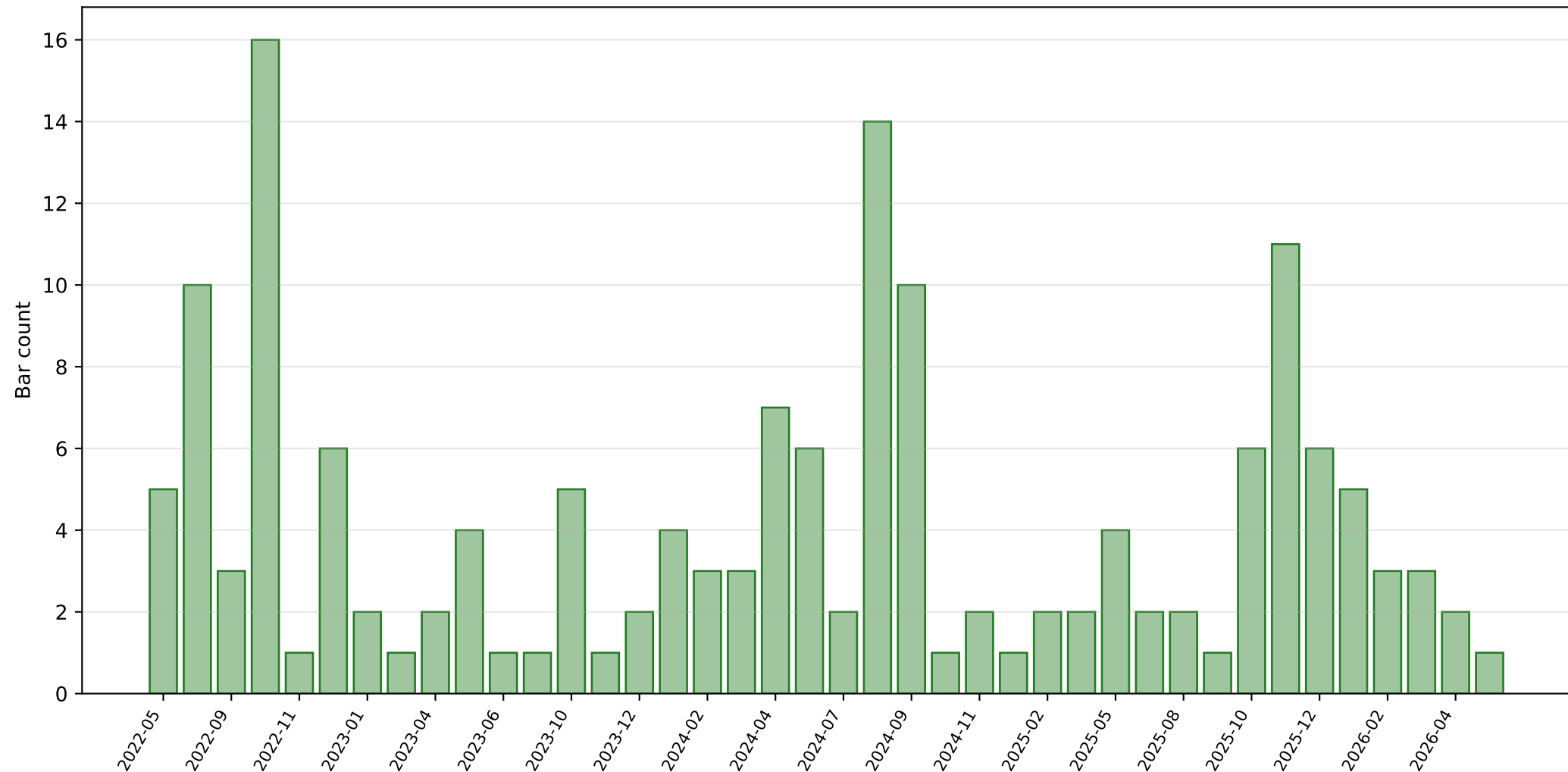
[narrow_range_break_fake]
Narrow range: range < prior min range over 8 bars; fade: short on bearish close, long on bullish close

[tiny_drift]
|ema_fast-ema_slow|/ATR < 0.2 (ema_fast=12, ema_slow=50, ATR period=14)
Long: 1-bar return > 0; short: 1-bar return < 0

Regime 45 — bar counts by year (UTC)
EURUSD · 60-minute bars (H1) · n=25,000 bars



Regime 45 — bar counts by month (UTC, last 40 months)
EURUSD · 60-minute bars (H1) · n=25,000 bars



Top strategies by rank_score — regime 45
EURUSD · 60-minute bars (H1) · n=25,000 bars

